

Thinh Ong

Infectious disease modelling and vaccine epidemiology

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RESEARCH INTERESTS

I model the spread of vaccine-preventable diseases in Vietnam, and help Ho Chi Minh City CDC build the surveillance and modelling capacity behind its outbreak response.

EDUCATION

2022-2026 **PhD**, University of Oxford, United Kingdom

2012-2018 **MD**, University of Medicine and Pharmacy at Ho Chi Minh City, Vietnam

EXPERIENCE

2021-2022 **Research assistant**, Mathematical Modelling Group, Oxford University Clinical Research Unit, Vietnam

2020-2021 **Editorial assistant**, MedPharmRes, University of Medicine and Pharmacy at Ho Chi Minh City, Vietnam

2019-2020 **Research intern**, Cancer Epigenetics Lab, Garvan Institute of Medical Research, Australia

2018-2023 **Biostatistical consultant**, Infertility Department, Tu Du Hospital, Vietnam

GRANTS

Secured \$237,930 in research grants since 2022.

2025-2026 *PI* \$55,456 **Understanding how immunisation inequities drive measles outbreaks.** University of Oxford's internal 'Improving Equitable Access to Healthcare Pump Priming'.

2024-2025 *PI* \$27,694 **Modelling measles and hand-foot-mouth disease transmission dynamics in Ho Chi Minh city.** OUCRU's Programme Scientific Committee.

2023-2025 *PI* \$29,961 **The impacts of 15-month disruption in childhood vaccination programme on vaccination coverage and seroprevalence in Ho Chi Minh city.** OUCRU's Programme Scientific Committee.

2022-2024	<i>PI</i>	\$20,437	An R package for modelling infectious disease parameters based on serological and social contact data. OUCRU's Programme Scientific Committee.
2024-2025	<i>Co-I</i>	\$104,382	Research cultures in Southeast Asia. Wellcome.

AWARDS

2022-2026	PhD Studentship: OUCRU & NDM Tropical Network Fund (University of Oxford)
2019	Outstanding Presentation , 36th Science & Technology Conference, Ho Chi Minh City, Vietnam
2018	Highest GPA in the 6-year medical doctor programme (highest in school history)
2018	Highest dissertation score
2013-2018	Annual highest GPA award

RESEARCH TEAM

<i>Tenure</i>	<i>Role</i>	<i>Name</i>	<i>Next destination</i>
2024-2026	RA	Pham Nguyen The Nguyen	MPH programme at National Cancer Center Graduate School of Cancer Science and Policy
2023-2026	RA	Phan Truong Quynh Anh	PhD programme at University of Oxford
2022-2023	RA	Huynh Ngoc Tuyen	PhD programme at University of Oxford
2024	Intern	Nguyen Thanh Lam	MPH programme at University of Caen Normandie
2024	Intern	Julian Hough	BSc programme at Princeton University
2024	Intern	Huynh Xuan Loc	PhD programme at Boston University

POLICY ENGAGEMENT

2026	Technical advisor to Ho Chi Minh City CDC on the city's Infectious Disease Early Warning and Response System
2024	Modelled measles seroprevalence and immunity gaps to inform the Department of Health's decision to declare a measles outbreak and design the reactive vaccination campaign. Recognised as one of the city's <u>top ten medical achievements of 2024</u>
2024	Advised the development of a dashboard for monitoring infectious diseases at Ho Chi Minh City CDC

2024-2026	Trained an in-house modelling team at Ho Chi Minh City CDC, now able to conduct research and run routine outbreak analyses independently
2024-2025	Advised on the evaluation of Ho Chi Minh City's Infectious Disease Surveillance System
2024-2025	Benchmarked and recommended epidemic-threshold detection algorithms for Ho Chi Minh City CDC

TEACHING

<i>Year</i>	<i>People</i>	<i>Course</i>
<i>Modelling</i>		
2025	30	Modelling and Analysis of Serological Data: An Introductory Short Course Teaching assistant for a course jointly developed by MRC Centre for Global Infectious Disease Analysis, Imperial College London and National University of Singapore. Oxford University Clinical Research Unit.
2024-2025	15	Infectious disease modelling for babies Methodologies for mathematical models of infectious diseases. HCMC's Center for Disease Control. drthinhong.com/idm4b
2023	60	Introduction to mathematical models for controlling infectious diseases Developing basic compartmental models and the use of reproduction numbers. HCMC's Center for Disease Control. lampk.github.io/IDMIntro
<i>Other courses</i>		
2025	20	Academic writing for babies Academic writing skills for early-career researchers. HCMC's Center for Disease Control.
2024-2025	10	R café Fortnightly reproducibility in research classes. Oxford University Clinical Research Unit, HCMC's Center for Disease Control. oucru-modelling.github.io/R-cafe-materials
2024	40	R for babies Monthly practical R programming classes. HCMC's Center for Disease Control. drthinhong.com/r4babies
<i>Seminars</i>		
2024	40	Is this a good journal? Training on identifying trustworthy scientific journals. HCMC's Center for Disease Control.

SOFTWARE

- 2025 **denim**. Construct and simulate deterministic discrete-time compartmental models with customisable dwell time distributions. thinhhong.github.io/denim
- 2025 **serosv**. Analyse serological data and estimate infectious diseases parameters. oucru-modelling.github.io/serosv
- 2025 **tvmedg**. Causal mediation analysis using g-computation. causalepi.github.io/tvmedg
- 2021 **MethPanel**. Analyse multiplex bisulphite PCR sequencing to assess DNA methylation biomarker panels for disease detection. github.com/thinhong/MethPanel
- 2020 **hss**. Estimate sample size for health study designs. thinhhong.shinyapps.io/samplesize
- 2020 **Lifted**. Enhance genome conversion robustness during reference assembly updates. github.com/thinhong/Lifted

PUBLICATIONS

1. **Ong TP**, Phan ATQ, Ha LM, Choisy M. denim: An R package for deterministic compartmental models with flexible dwell-time distributions. *Methods in Ecology and Evolution*. 2026;17(4):1060-1068. doi:[10.1111/2041-210x.70256](https://doi.org/10.1111/2041-210x.70256)
2. Huynh TM, **Ong T**, Tran DT, Quach DT, Vo TD. Neutrophil-C-reactive protein index as a novel prognostic tool for acute pancreatitis in resource-constrained settings. *International Journal of Surgery*. 2026;112(2):3297. doi:[10.1097/JS9.00000000000003846](https://doi.org/10.1097/JS9.00000000000003846)
3. Nguyen QD, **Ong TP**, Truong LTT, Trinh TH, Le HN, Do LAH, Huynh J. High vulnerability of infants under 6 months of age in Vietnam's measles outbreak. *Vaccine*. 2026;77:128375. doi:[10.1016/j.vaccine.2026.128375](https://doi.org/10.1016/j.vaccine.2026.128375)
4. Huynh TM, **Ong T**. PancreaSure: Ready for Real-World PDAC Surveillance or Still Limited to a Highly Selected Cohort?. *Gastroenterology*. 2026. doi:[10.1053/j.gastro.2026.01.049](https://doi.org/10.1053/j.gastro.2026.01.049)
5. Huynh TM, **Ong T**, Gautam N, Nguyen HS. Refining intergenerational MASLD risk: are measurement and mediation shaping the signal?. *Gut*. 2026. doi:[10.1136/gutjnl-2026-338664](https://doi.org/10.1136/gutjnl-2026-338664)
6. Huynh T, **Ong T**. Are S100A7 protein and V-set and immunoglobulin domain-containing 4 specific and robust early markers of severe acute pancreatitis?. *Polish Archives of Internal Medicine*. 2026. doi:[10.20452/pamw.17238](https://doi.org/10.20452/pamw.17238)
7. **Ong T**, Thuy CT, Tam NH, Nga LH, Uyen NTV, Lan TTT, Tran A, Diem LTK, Khang HP, Dung DTX, Chuong LN, Tam NHT, Giang VC, Cam TTB, Anh HTN, Thu PTM, Man DNH, Dung NT, Tung NLN, Phat HH, Toan LM, Lam NT, Nguyen PNT, Nhung HN, Thanh TT, Chau NVV, Thuong TC, Thwaites G, Thwaites CL, Choisy M, Tan LV. Detection of Immunity Gap before Measles Outbreak, Ho Chi Minh City, Vietnam, 2024. *Emerging Infectious Diseases*. 2025;31(10). doi:[10.3201/eid3110.250234](https://doi.org/10.3201/eid3110.250234)

8. Phi HD, Ca TN, Oanh PM, Phong NT, **Ong T**, Nhung PTH. Impact of measles vaccination on disease severity and economic burden among children in northern Vietnam, 2017–2019. *Vaccine: X*. 2025;26:100714. doi:[10.1016/j.jvacx.2025.100714](https://doi.org/10.1016/j.jvacx.2025.100714)
9. Leung K, Cook AR, Wu JT, McVernon J, Prem K, Jit M, Kingkaew P, Mukhopadhyay S, Isaranuwatthai W, Pan-ngum W, Teerawattananon Y, Huang A, Djaafara B, Sukmanee J, **Ong T**, Laemlak S, Chua VW, Rattanavipapong W, Dabak SV. Enhancing global health security responses through greater inclusion of the global south in infectious disease modelling. *BMJ Glob Health*. 2025;10(8). doi:[10.1136/bmjgh-2025-019111](https://doi.org/10.1136/bmjgh-2025-019111)
10. Le TMC, Duong KT, **Ong T**, Nguyen THN, Thai TCT, Nguyen TTH, Nguyen QA, Le QT, Pham NTN, Roque M, Alviggi C, Conforti A. O-187 Individualized luteal phase support with intramuscular progesterone in patients with low serum progesterone on the day of frozen-thawed embryo transfer: a randomized clinical trial. *Hum Reprod*. 2025;40(Supplement_1):deaf097.187. doi:[10.1093/humrep/deaf097.187](https://doi.org/10.1093/humrep/deaf097.187)
11. Le TMC, Tran THD, Pham VP, Dang TL, Duong KT, Hua TT, Huynh TNT, Nguyen THN, Le QT, Nguyen BMN, Vo MT, **Ong T**, Pham NTN, Amorim CA. The first comparative study on the effectiveness of slow freezing and vitrification of human ovarian tissue by xenotransplantation model in Vietnam. *J Assist Reprod Genet*. 2025;42(5):1473-1484. doi:[10.1007/s10815-025-03439-z](https://doi.org/10.1007/s10815-025-03439-z)
12. Pham NML, **Ong T**, Vuong NL, Nguyen TTH. HLA Compatibility and Graft Survival Rates Among Related and Unrelated Donors in Renal Transplantation. *Transplantation Proceedings*. 2024;56(10):2163-2171. doi:[10.1016/j.transproceed.2024.11.010](https://doi.org/10.1016/j.transproceed.2024.11.010)
13. Tran DM, **Ong T**, Cao TV, Pham QT, Do H, Phan PH, Choisy M, Pham NTH. Hospital-acquired infections and unvaccinated children due to chronic diseases: an investigation of the 2017–2019 measles outbreak in the northern region of Vietnam. *BMC Infect Dis*. 2024;24(1):948. doi:[10.1186/s12879-024-09816-w](https://doi.org/10.1186/s12879-024-09816-w)
14. Le Pham NM, **Ong T**, Vuong NL, Van Tran B, Nguyen TTH. HLA types and their association with end-stage renal disease in Vietnamese patients: A cross-sectional study. *Medicine*. 2022;101(48):e31856. doi:[10.1097/MD.00000000000031856](https://doi.org/10.1097/MD.00000000000031856)
15. Le TMC, **Ong T**, Nguyen QA, Roque M. Fresh versus elective frozen embryo transfer: Cumulative live birth rates of 7,236 IVF cycles. *JBRA Assist Reprod*. 2022;26(3):450-459. doi:[10.5935/1518-0557.20210094](https://doi.org/10.5935/1518-0557.20210094)
16. Le TTV, Vuong TTB, **Ong T**, Do MD. Allele frequency and the associations of HLA-DRB1 and HLA-DQB1 polymorphisms with pemphigus subtypes and disease severity. *Medicine*. 2022;101(7):e28855. doi:[10.1097/MD.00000000000028855](https://doi.org/10.1097/MD.00000000000028855)
17. Le TMC, Duong KT, Nguyen QA, **Ong T**, Nguyen THN, Thai TCT, Le QT, Roque M, Alviggi C. Effectiveness of progesterone supplementation in women presenting low progesterone levels on the day of frozen embryo transfer: a randomised controlled trial. *BMJ Open*. 2022;12(2):e057353. doi:[10.1136/bmjopen-2021-057353](https://doi.org/10.1136/bmjopen-2021-057353)

18. Long KQ, Ngoc-Anh HT, Phuong NH, Tuyet-Hanh TT, Park K, Takeuchi M, Lam NT, Nga PTQ, Phuong-Anh L, Tuan LV, Bao TQ, **Ong T**, Huy NV, Lan VTH, Minh HV. Clustering Lifestyle Risk Behaviors among Vietnamese Adolescents and Roles of School: A Bayesian Multilevel Analysis of Global School-Based Student Health Survey 2019. *The Lancet Regional Health – Western Pacific*. 2021;15:100225. doi:[10.1016/j.lanwpc.2021.100225](https://doi.org/10.1016/j.lanwpc.2021.100225)
19. Luu PL, **Ong T**, Loc TTH, Lam D, Pidsley R, Stirzaker C, Clark SJ. MethPanel: a parallel pipeline and interactive analysis tool for multiplex bisulphite PCR sequencing to assess DNA methylation biomarker panels for disease detection. *Bioinformatics*. 2021;37(15):2198-2200. doi:[10.1093/bioinformatics/btaa1060](https://doi.org/10.1093/bioinformatics/btaa1060)
20. Vuong TBT, Do DM, **Ong T**, Thanh Le TV. HLA-DRB1 and DQB1 genetic susceptibility to pemphigus vulgaris and pemphigus foliaceus in Vietnamese patients. *Dermatol Reports*. 2021;14(2):9286. doi:[10.4081/dr.2021.9286](https://doi.org/10.4081/dr.2021.9286)
21. Huynh G, Tran TT, Do THT, Truong TTD, **Ong T**, Nguyen TNH, Pham LA. Diabetes-Related Distress Among People with Type 2 Diabetes in Ho Chi Minh City, Vietnam: Prevalence and Associated Factors. *Diabetes, Metabolic Syndrome and Obesity*. 2021;14:683-690. doi:[10.2147/DMSO.S297315](https://doi.org/10.2147/DMSO.S297315)
22. Long KQ, **Ong T**, Thao TTK, Van Huy N, Lan VTH, Mai VQ, Van Minh H. Relationship between family functioning and health-related quality of life among methadone maintenance patients: a Bayesian approach. *Qual Life Res*. 2020;29(12):3333-3342. doi:[10.1007/s11136-020-02598-z](https://doi.org/10.1007/s11136-020-02598-z)
23. Luu PL, **Ong T**, Dinh TP, Clark SJ. Benchmark study comparing liftover tools for genome conversion of epigenome sequencing data. *NAR Genomics and Bioinformatics*. 2020;2(3):lqaa054. doi:[10.1093/nargab/lqaa054](https://doi.org/10.1093/nargab/lqaa054)
24. Vu-Ngoc H, Uyen NCM, **Ong T**, Don LD, Danh NVT, Truc NTT, Vi VT, Vuong NL, Huy NT, Duong PDT. Analgesic effect of non-nutritive sucking in term neonates: A randomized controlled trial. *Pediatrics & Neonatology*. 2020;61(1):106-113. doi:[10.1016/j.pedneo.2019.07.003](https://doi.org/10.1016/j.pedneo.2019.07.003)
25. Dao Phuoc T, Khuong Quynh L, Vien Dang Khanh L, **Ong T**, Le Sy H, Le Ngoc T, Phung Khanh L. Clinical prognostic models for severe dengue: a systematic review protocol. *Wellcome Open Res*. 2019;4:12. doi:[10.12688/wellcomeopenres.15033.2](https://doi.org/10.12688/wellcomeopenres.15033.2)
26. **Ong T**, Anh HNV, Tung DT, Kien TG. Translation and cross-cultural adaptation of the Vietnamese version of the diabetes distress scale. *MedPharmRes*. 2018;2(3):5-11. doi:[10.32895/UMP.MPR.2.3.5](https://doi.org/10.32895/UMP.MPR.2.3.5)
27. Huynh TM, Gautam N, **Ong T**. Is the selenium paradox in NAFLD a result of reverse causation and nutritional confounding?. *International Journal of Surgery*. ;10.1097/JS9.0000000000005190. doi:[10.1097/JS9.0000000000005190](https://doi.org/10.1097/JS9.0000000000005190)